

I. Kết Nối SSH & Bảo Mật

Test kết nối key:

```
PS C:\Users\hungdg42304> ssh hungdg@172.18.236.53
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.114.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jun 16 15:05:58 +07 2026

System load:  0.0      Processes:            31
Usage of /:   0.2% of 1006.85GB Users logged in:      0
Memory usage: 5%      IPv4 address for eth0: 172.18.236.53
Swap usage:   0%

Last login: Tue Jun 16 14:54:00 2026 from 172.18.224.1
hungdg@PC-BNKBJ167:~$
```

Sau khi cấu hình

```
Port 2222
PermitRootLogin no
PasswordAuthentication no
PubkeyAuthentication yes
MaxAuthTries 3
ClientAliveInterval 300
ClientAliveCountMax 3
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
```

Test kết nối port 2222

```
PS C:\Users\hungdg42304> ssh hungdg@172.18.236.53 -p 2222
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.114.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jun 16 15:08:42 +07 2026

System load:  0.06      Processes:            31
Usage of /:   0.2% of 1006.85GB Users logged in:      0
Memory usage: 5%      IPv4 address for eth0: 172.18.236.53
Swap usage:   0%

Last login: Tue Jun 16 15:05:58 2026 from 172.18.224.1
```

Lỗi permission thư mục /.ssh

```
ssh hungdg@172.18.236.53 -p 2222
Permission denied (publickey).
```

Check log lỗi

```
ungdg@PC-BNKB167:~$ sudo tail -f /var/log/auth.log
026-06-16T14:30:02.073786+07:00 PC-BNKB167 sudo: hungdg
bin/systemctl restart sshd
```

```
Could not open user 'hungdg' authorized keys '/home/hungdg/.ssh/authorized_keys': Permission denied
Could not open user 'hungdg' authorized keys '/home/hungdg/.ssh/authorized_keys': Permission denied
Connection reset by authenticating user hungdg 172.18.224.1 port 50064 [preauth]
:session): session opened for user root(uid=0) by (uid=1000)
```

Phân quyền thư mục

```
hungdg@PC-BNKB167:~$ sudo chown -R hungdg:hungdg /home/hungdg/.ssh
hungdg@PC-BNKB167:~$ sudo chmod 700 /home/hungdg/.ssh
hungdg@PC-BNKB167:~$ sudo chmod 600 /home/hungdg/.ssh/authorized_keys
hungdg@PC-BNKB167:~$ sudo chmod 755 /home/hungdg
hungdg@PC-BNKB167:~$ ls -ld /home/hungdg
drwxr-xr-x 5 hungdg hungdg 4096 Jun 16 13:51 /home/hungdg
hungdg@PC-BNKB167:~$ ls -ld /home/hungdg/.ssh
drwx----- 2 hungdg hungdg 4096 Jun 16 14:29 /home/hungdg/.ssh
hungdg@PC-BNKB167:~$ ls -l /home/hungdg/.ssh
total 12
-rw----- 1 hungdg hungdg 115 Jun 16 14:29 authorized_keys
-rw----- 1 hungdg hungdg 411 Jun 16 13:51 id_ed25519
-rw-r--r-- 1 hungdg hungdg 100 Jun 16 13:51 id_ed25519.pub
hungdg@PC-BNKB167:~$ sudo systemctl restart sshd
```

Cấu hình SSH Tunneling & SCP

Chuyển file từ win → linux

```
PS C:\Users\hungdg42304> scp -i $HOME\.ssh\id_ed25519 -P 2222 `
>> "C:\Users\hungdg42304\test.txt" `
>> hungdg@172.18.236.53:/home/hungdg/
test.txt
100% 74 0.1KB/s 00:00
```

```
hungdg@PC-BNKB167:~$ ls -l /home/hungdg/test.txt
-rw-rw-r-- 1 hungdg hungdg 74 Jun 16 15:43 /home/hungdg/test.txt
hungdg@PC-BNKB167:~$ cat /home/hungdg/test.txt
Test file chuyển scp từ windows qua Linux ---- Thành công -----hungdg@PC-BNKB167:~$
```

Chuyển folder từ win → linux

```
PS C:\Users\hungdg42304> scp -r -i $HOME\.ssh\id_ed25519 -P 2222 `
>> "C:\Users\hungdg42304\project" `
>> hungdg@172.18.236.53:/home/hungdg/
test2.txt 100% 74 0.1KB/s 00:00
```

```
hungdg@PC-BNKB167:~$ tree /home/hungdg/project
/home/hungdg/project
└── test2.txt

1 directory, 1 file
hungdg@PC-BNKB167:~$
```

SSH Local Port Forwarding

Flow thực tế

Browser Windows

↓

localhost:8080

↓

SSH encrypted tunnel

↓

WSL Ubuntu localhost:80

```
PS C:\Users\hungdg42304> ssh -p 2222 -L 8080:localhost:80 hungdg@172.18.236.53
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.114.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jun 16 16:05:19 +07 2026

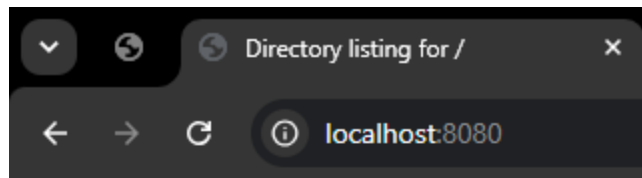
System load:  0.0               Processes:    34
Usage of /:   0.2% of 1006.85GB  Users logged in: 0
Memory usage: 5%               IPv4 address for eth0: 172.18.236.53
Swap usage:   0%

Last login: Tue Jun 16 15:08:42 2026 from 172.18.224.1
```

Test chạy thử 1 webserver trên port 80

```
hungdg@PC-BNKBJ167:~$ sudo python3 -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
127.0.0.1 - - [16/Jun/2026 16:09:25] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [16/Jun/2026 16:09:25] code 404, message File not found
127.0.0.1 - - [16/Jun/2026 16:09:25] "GET /favicon.ico HTTP/1.1" 404 -
```

Trên windows truy cập port http://localhost:8080



Directory listing for /

- [.bash_history](#)
- [.bash_logout](#)
- [.bashrc](#)
- [.cache/](#)
- [.config/](#)
- [.motd_shown](#)
- [.profile](#)
- [.selected_editor](#)
- [.ssh/](#)
- [cron.log](#)
- [project/](#)
- [test.sh](#)
- [test.txt](#)

Ví dụ:

Flow Jenkins

Browser Windows



localhost:9999



SSH tunnel

↓

Linux Jenkins :8080

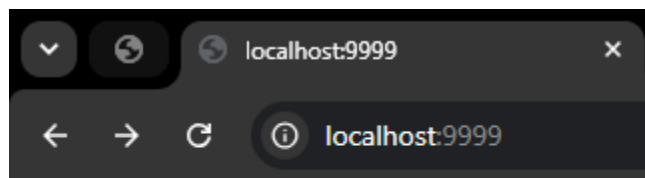
Chạy server web tạm Trên linux

```
hungdg@PC-BNKBJ167:~$ mkdir -p ~/demo
hungdg@PC-BNKBJ167:~$ cd ~/demo
hungdg@PC-BNKBJ167:~/demo$ echo hello tunnel > index.html
hungdg@PC-BNKBJ167:~/demo$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

Tạo tunnel từ Windows

```
PS C:\Users\hungdg42304> ssh -p 2222 -L 9999:localhost:8080 hungdg@172.18.236.53
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.114.1-microsoft-standard-WSL2 x86_64)
```

Test <http://localhost:9999>



hello tunnel

Chạy lệnh từ xa:

```
PS C:\Users\hungdg42304> ssh -p 2222 hungdg@172.18.236.53 "pwd && whoami && uptime"
/home/hungdg
hungdg
16:31:12 up 8:39, 2 users, load average: 0.00, 0.00, 0.00

PS C:\Users\hungdg42304> ssh -p 2222 hungdg@172.18.236.53 "ls -l /home/hungdg"
total 40
-rw-r--r-- 1 hungdg hungdg 20304 Jun 16 16:30 cron.log
drwxr-xr-x 2 hungdg hungdg 4096 Jun 16 16:18 demo
drwx----- 2 hungdg hungdg 4096 Jun 16 15:58 project
-rwxr-xr-x 1 root root 72 Jun 16 08:08 test.sh
-rw-rw-r-- 1 hungdg hungdg 74 Jun 16 15:48 test.txt
```

```
PS C:\Users\hungdg42304> ssh -p 2222 hungdg@172.18.236.53 "df -h"
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G   0    3.9G   0% /usr/lib/modules/6.6.114.1-microsoft-standard-WSL2
none            3.9G  4.0K   3.9G   1% /mnt/wsl
drivers          477G  207G  270G  44% /usr/lib/wsl/drivers
/dev/sdd        1007G   1.6G  955G   1% /
none            3.9G   80K   3.9G   1% /mnt/wslg
none            3.9G   0    3.9G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.8M   3.8G   1% /init
none            3.9G  544K   3.9G   1% /run
none            3.9G   0    3.9G   0% /run/lock
none            3.9G   0    3.9G   0% /run/shm
none            3.9G   80K   3.9G   1% /mnt/wslg/versions.txt
none            3.9G   80K   3.9G   1% /mnt/wslg/doc
C:\              477G  207G  270G  44% /mnt/c
D:\              466G   53G  413G  12% /mnt/d
none            1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           3.9G  4.0K   3.9G   1% /tmp
none            1.0M   0    1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           779M  12K   779M   1% /run/user/1000
```

```
PS C:\Users\hungdg42304> ssh -p 2222 hungdg@172.18.236.53 "touch demo.txt"
PS C:\Users\hungdg42304> ssh -p 2222 hungdg@172.18.236.53 "echo hello > demo.txt"
PS C:\Users\hungdg42304> ssh -p 2222 hungdg@172.18.236.53 "ls -l /home/hungdg"
total 44
-rw-r--r-- 1 hungdg hungdg 20448 Jun 16 16:33 cron.log
drwxr-xr-x 2 hungdg hungdg 4096 Jun 16 16:18 demo
-rw-rw-r-- 1 hungdg hungdg 6 Jun 16 16:32 demo.txt
drwx----- 2 hungdg hungdg 4096 Jun 16 15:58 project
-rwxr-xr-x 1 root root 72 Jun 16 08:08 test.sh
-rw-rw-r-- 1 hungdg hungdg 74 Jun 16 15:48 test.txt
PS C:\Users\hungdg42304> ssh -p 2222 hungdg@172.18.236.53 "cat /home/hungdg/demo.txt"
hello
```

II. CRONJOB

Thiết lập thư mục + file cấu hình

```
hungdg@PC-BNKB167:~$ sudo mkdir -p /home/hungdg/lab01/scripts
[sudo: authenticate] Password:
hungdg@PC-BNKB167:~$ ls -l -p /home/hungdg/lab01/scripts
total 0
hungdg@PC-BNKB167:~$ ls -l -p /home/hungdg/lab01
total 4
drwxr-xr-x 2 hungdg hungdg 4096 Jun 17 08:08 scripts/
hungdg@PC-BNKB167:~$ ls -l -p /home/hungdg
total 52
-rw-r--r-- 1 hungdg hungdg 22608 Jun 17 08:09 cron.log
drwxr-xr-x 2 hungdg hungdg 4096 Jun 16 16:18 demo/
-rw-rw-r-- 1 hungdg hungdg 6 Jun 16 16:32 demo.txt
drwxr-xr-x 3 hungdg hungdg 4096 Jun 17 08:08 lab01/
drwx----- 2 hungdg hungdg 4096 Jun 16 15:58 project/
-rwxr-xr-x 1 root root 72 Jun 16 08:08 test.sh
-rw-rw-r-- 1 hungdg hungdg 74 Jun 16 15:48 test.txt
```

Phân quyền + add .log

```
hungdg@PC-BNKB167:~$ sudo vim /home/hungdg/lab01/scripts/syscheck.sh
hungdg@PC-BNKB167:~$ sudo chmod 755 /home/hungdg/lab01/scripts/syscheck.sh
hungdg@PC-BNKB167:~$ sudo ls -l /home/hungdg/lab01/scripts/syscheck.sh
-rwxr-xr-x 1 root root 69 Jun 17 08:13 /home/hungdg/lab01/scripts/syscheck.sh
hungdg@PC-BNKB167:~$ sudo cat /home/hungdg/lab01/scripts/syscheck.sh
#!/bin/bash

echo "cronjob running at $(date)" >> /tmp/syscheck.log
```

```
hungdg@PC-BNKB167:/$ sudo vim /home/hungdg/lab01/scripts/backup.sh
hungdg@PC-BNKB167:/$ sudo chmod 755 /home/hungdg/lab01/scripts/backup.sh
hungdg@PC-BNKB167:/$ sudo ls -l /home/hungdg/lab01/scripts/backup.sh
-rwxr-xr-x 1 root root 66 Jun 17 08:22 /home/hungdg/lab01/scripts/backup.sh
hungdg@PC-BNKB167:/$ sudo cat /home/hungdg/lab01/scripts/backup.sh
#!/bin/bash

echo "cronjob running at $(date)" >> /tmp/backup.log
hungdg@PC-BNKB167:/$
```

Mở crontab

```
hungdg@PC-BNKB167:/$ sudo crontab -e
```

```
# Chạy script kiểm tra hệ thống mỗi 5 phút, lưu log
*/5 * * * * /home/hungdg/lab01/scripts/syscheck.sh >> /tmp/syscheck.log 2>&1

# Backup hàng ngày lúc 2 giờ sáng
0 2 * * * /home/hungdg/lab01/scripts/backup.sh >> /tmp/backup.log 2>&1

# Xóa file /tmp cũ hơn 7 ngày, mỗi chủ nhật
0 3 * * 0 find /tmp -mtime +7 -delete 2>/dev/null
```

Check log

```
hungdg@PC-BNKB167:/$ sudo cat /tmp/syscheck.log
cronjob running at Wed Jun 17 14:45:01 +07 2026
cronjob running at Wed Jun 17 14:50:01 +07 2026
cronjob running at Wed Jun 17 14:55:01 +07 2026
cronjob running at Wed Jun 17 15:00:01 +07 2026
cronjob running at Wed Jun 17 15:05:01 +07 2026
cronjob running at Wed Jun 17 15:10:01 +07 2026
cronjob running at Wed Jun 17 15:15:01 +07 2026
cronjob running at Wed Jun 17 15:20:01 +07 2026
cronjob running at Wed Jun 17 15:25:01 +07 2026
cronjob running at Wed Jun 17 15:30:02 +07 2026
```

```
hungdg@PC-BNKB167:/$ sudo grep cron /tmp/syscheck.log | tail -20
cronjob running at Wed Jun 17 14:45:01 +07 2026
cronjob running at Wed Jun 17 14:50:01 +07 2026
cronjob running at Wed Jun 17 14:55:01 +07 2026
cronjob running at Wed Jun 17 15:00:01 +07 2026
cronjob running at Wed Jun 17 15:05:01 +07 2026
cronjob running at Wed Jun 17 15:10:01 +07 2026
cronjob running at Wed Jun 17 15:15:01 +07 2026
cronjob running at Wed Jun 17 15:20:01 +07 2026
cronjob running at Wed Jun 17 15:25:01 +07 2026
cronjob running at Wed Jun 17 15:30:02 +07 2026
cronjob running at Wed Jun 17 15:35:01 +07 2026
```

```
hungdg@PC-BNKB167:~$ sudo grep cron /tmp/backup.log | tail -20
[sudo: authenticate] Password:
cronjob running at Thu Jun 18 15:00:01 +07 2026
cronjob running at Thu Jun 18 16:00:01 +07 2026
```


Kiểm tra

```
hungdg@PC-BNKB167:/$ echo 'echo test >> /tmp/at_test2.txt' | at now + 2 minute
warning: commands will be executed using /bin/sh
job 10 at Wed Jun 17 16:08:00 2026
hungdg@PC-BNKB167:/$ ls -l /tmp
total 12
-rw-r--r-- 1 hungdg hungdg 10 Jun 17 15:06 at_test.txt
drwx----- 2 root crontab 60 Jun 17 15:48 crontab.kS6nIH
-rw-r--r-- 1 hungdg hungdg 6 Jun 17 16:02 hello.txt
drwx----- 2 root root 40 Jun 17 08:02 snap-private-tmp
-rw-r--r-- 1 root root 816 Jun 17 16:05 syscheck.log
drwx----- 3 root root 60 Jun 17 08:02 systemd-private-8eb5d9b43ca148e6b98ef03484315084-chrony.service-0vjaJ1
drwx----- 3 root root 60 Jun 17 14:54 systemd-private-8eb5d9b43ca148e6b98ef03484315084-polkit.service-F00Dkg
drwx----- 3 root root 60 Jun 17 08:02 systemd-private-8eb5d9b43ca148e6b98ef03484315084-systemd-logind.service-6EI9YY
-rw-r--r-- 1 hungdg hungdg 0 Jun 17 16:04 test_at
hungdg@PC-BNKB167:/$ atq
10 Wed Jun 17 16:08:00 2026 a hungdg
```

```
hungdg@PC-BNKB167:/$ ls -l /tmp
total 16
-rw-r--r-- 1 hungdg hungdg 15 Jun 17 16:06 at_test.txt
-rw-r--r-- 1 hungdg hungdg 5 Jun 17 16:08 at_test2.txt
drwx----- 2 root crontab 60 Jun 17 15:48 crontab.kS6nIH
-rw-r--r-- 1 hungdg hungdg 6 Jun 17 16:02 hello.txt
drwx----- 2 root root 40 Jun 17 08:02 snap-private-tmp
-rw-r--r-- 1 root root 864 Jun 17 16:10 syscheck.log
drwx----- 3 root root 60 Jun 17 08:02 systemd-private-8eb5d9b43ca148e6b98ef03484315084-chrony.service-0vjaJ1
drwx----- 3 root root 60 Jun 17 14:54 systemd-private-8eb5d9b43ca148e6b98ef03484315084-polkit.service-F00Dkg
drwx----- 3 root root 60 Jun 17 08:02 systemd-private-8eb5d9b43ca148e6b98ef03484315084-systemd-logind.service-6EI9YY
-rw-r--r-- 1 hungdg hungdg 0 Jun 17 16:04 test_at
```

```
hungdg@PC-BNKB167:/$ atq
hungdg@PC-BNKB167:/$
```

III. Quản lý và phân quyền user

Tạo user group:

```
hungdg@PC-BNKB167:/$ sudo useradd -m -s /bin/bash -G developers alice
hungdg@PC-BNKB167:/$ sudo useradd -m -s /bin/bash -G developers,devops bob
hungdg@PC-BNKB167:/$ sudo passwd alice
New password:
Retype new password:
passwd: password updated successfully
hungdg@PC-BNKB167:/$ sudo passwd bob
New password:
Retype new password:
passwd: password updated successfully
hungdg@PC-BNKB167:/$ id alice
uid=1001(alice) gid=1003(alice) groups=1003(alice),1001(developers)
```

Check user và groups + add user vào group

```
hungdg@PC-BNKB167:/$ cat /etc/passwd | grep -E "alice|bob"
alice:x:1001:1003:~/home/alice:/bin/bash
bob:x:1002:1004:~/home/bob:/bin/bash
hungdg@PC-BNKB167:/$ sudo usermod -aG devops alice
hungdg@PC-BNKB167:/$ id alice
uid=1001(alice) gid=1003(alice) groups=1003(alice),1001(developers),1002(devops)
hungdg@PC-BNKB167:/$ |
```

```
hungdg@PC-BNKB167:~$ groups alice
alice : alice developers devops
hungdg@PC-BNKB167:~$ groups bob
bob : bob developers devops
hungdg@PC-BNKB167:~$ |
```

Tạo nhiều folder 1 lúc

```
hungdg@PC-BNKB167:/$ sudo mkdir -p /opt/project/{shared,dev,deploy}
hungdg@PC-BNKB167:/$ sudo ls -l /opt/project
total 12
drwxr-xr-x 2 root root 4096 Jun 18 10:04 deploy
drwxr-xr-x 2 root root 4096 Jun 18 10:04 dev
drwxr-xr-x 2 root root 4096 Jun 18 10:04 shared
```

Phân quyền user cho từng thư mục

```
hungdg@PC-BNKB167:/$ sudo chown root:developers /opt/project/dev
hungdg@PC-BNKB167:/$ sudo chmod 770 /opt/project/dev
```

```
hungdg@PC-BNKB167:/$ sudo chown root:devops /opt/project/deploy
hungdg@PC-BNKB167:/$ sudo chmod 750 /opt/project/deploy
```

```
hungdg@PC-BNKB167:/$ sudo chown root:developers /opt/project/shared
hungdg@PC-BNKB167:/$ sudo chmod 1775 /opt/project/shared
```

```
hungdg@PC-BNKB167:/$ sudo ls -la /opt/project
total 20
drwxr-xr-x 5 root root      4096 Jun 18 10:04 .
drwxr-xr-x 3 root root      4096 Jun 18 10:04 ..
drwxr-x--- 2 root devops    4096 Jun 18 10:04 deploy
drwxrwx--- 2 root developers 4096 Jun 18 10:04 dev
drwxrwxr-t 2 root developers 4096 Jun 18 10:04 shared
```

Đăng nhập user tạo thư mục test:

```
hungdg@PC-BNKB167:/$ sudo -u alice touch /opt/project/dev/test_alice.txt
hungdg@PC-BNKB167:/$ sudo -u bob touch /opt/project/shared/test_bob.txt
```

Test sticky bit: user bob ko xóa được file của alice (**chỗ này trong đề hướng dẫn sai hoặc lừa mấy đứa gà !!!**)

```
hungdg@PC-BNKB167:/$ sudo -u bob touch /opt/project/dev/test_alice.txt
touch: setting times of '/opt/project/dev/test_alice.txt': Permission denied
```

Làm lại cho đúng:

```
hungdg@PC-BNKB167:~$ sudo -u alice touch /opt/project/shared/test2_alice.txt
hungdg@PC-BNKB167:~$ sudo -u bob rm /opt/project/shared/test2_alice.txt
rm: remove write-protected regular empty file '/opt/project/shared/test2_alice.txt'? yes
rm: cannot remove '/opt/project/shared/test2_alice.txt': Operation not permitted
hungdg@PC-BNKB167:~$ sudo -u alice rm /opt/project/shared/test2_alice.txt
hungdg@PC-BNKB167:~$ sudo cat /opt/project/shared/test2_alice.txt
cat: /opt/project/shared/test2_alice.txt: No such file or directory
hungdg@PC-BNKB167:~$ |
```

Chỉ alice mới xóa đc file của mình:

```
hungdg@PC-BNKB167:/$ sudo -u alice rm /opt/project/dev/test_alice.txt
hungdg@PC-BNKB167:/$ sudo -u alice ls -l /opt/project/dev/test_alice.txt
ls: cannot access '/opt/project/dev/test_alice.txt': No such file or directory
hungdg@PC-BNKB167:/$ sudo -u alice ls -l /opt/project/dev/
total 0
```

Chỉ cho alice có quyền sudo systemctl restart

```
hungdg@PC-BNKB167:~$ sudo visudo -f /etc/sudoers.d/alice
```

```
alice ALL=(ALL) NOPASSWD: /bin/systemctl restart nginx
alice ALL=(ALL) NOPASSWD: /bin/systemctl status *
```

Kiểm tra

```
hungdg@PC-BNKB167:~$ sudo -u alice sudo systemctl status sshd
o ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:sshd_config(5)
hungdg@PC-BNKB167:~$ sudo -u alice sudo apt install vim
sudo: I'm sorry alice. I'm afraid I can't do that
```

IV. Server thực tế

Chuẩn bị hệ thống

Cài ufw

```
hungdg@PC-BNKB167:~$ sudo apt install ufw -y
Installing:
  ufw
```

```
hungdg@PC-BNKB167:~$ ufw --version
ufw 0.36.2
Copyright 2008-2023 Canonical Ltd.
```

```
hungdg@PC-BNKB167:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
```

To	Action	From
--	-----	----
2222/tcp	ALLOW IN	Anywhere
80/tcp	ALLOW IN	Anywhere
443/tcp	ALLOW IN	Anywhere
2222/tcp (v6)	ALLOW IN	Anywhere (v6)
80/tcp (v6)	ALLOW IN	Anywhere (v6)
443/tcp (v6)	ALLOW IN	Anywhere (v6)

Cấu hình use group

```
hungdg@PC-BNKB167:~$ for user in alice bob carol; do
> sudo useradd -m -s /bin/bash -G webdev $user
  sudo mkdir -p /home/$user/.ssh
  sudo chmod 700 /home/$user/.ssh
  echo '# Paste public key here' | sudo tee /home/$user/.ssh/authorized_keys
  sudo chmod 600 /home/$user/.ssh/authorized_keys
  sudo chown -R $user:$user /home/$user/.ssh
  echo "User $user đã được tạo"
> done
# Paste public key here
User alice đã được tạo
# Paste public key here
User bob đã được tạo
# Paste public key here
User carol đã được tạo
```

```

hungdg@PC-BNKB167:~$ sudo tee /etc/sudoers.d/webdev << 'EOF'
> %webdev ALL=(ALL) NOPASSWD: /bin/systemctl restart nginx
> %webdev ALL=(ALL) NOPASSWD: /bin/systemctl reload nginx
> %webdev ALL=(ALL) NOPASSWD: /usr/bin/tail -f /var/log/nginx/*
> EOF
/etc/sudoers.d/webdev:3:29: wildcards are not allowed in command arguments
%webdev ALL=(ALL) NOPASSWD: /usr/bin/tail -f /var/log/nginx/*
^ ~~~~~
%webdev ALL=(ALL) NOPASSWD: /bin/systemctl restart nginx
%webdev ALL=(ALL) NOPASSWD: /bin/systemctl reload nginx
%webdev ALL=(ALL) NOPASSWD: /usr/bin/tail -f /var/log/nginx/*
hungdg@PC-BNKB167:~$ sudo vim /etc/sudoers.d/webdev
/etc/sudoers.d/webdev:3:29: wildcards are not allowed in command arguments
%webdev ALL=(ALL) NOPASSWD: /usr/bin/tail -f /var/log/nginx/*
^ ~~~~~

```

```

%webdev ALL=(ALL) NOPASSWD: /bin/systemctl restart nginx
%webdev ALL=(ALL) NOPASSWD: /bin/systemctl reload nginx
%webdev ALL=(ALL) NOPASSWD: /usr/bin/tail -f /var/log/nginx/access.log
%webdev ALL=(ALL) NOPASSWD: /usr/bin/tail -f /var/log/nginx/error.log

```

Test bằng user alice

```

alice@PC-BNKB167:~$ sudo -l
User alice may run the following commands on PC-BNKB167:
    (ALL) NOPASSWD: /usr/bin/systemctl restart nginx
    (ALL) NOPASSWD: /usr/bin/systemctl reload nginx
    (ALL) NOPASSWD: /usr/bin/tail -f /var/log/nginx/access.log
    (ALL) NOPASSWD: /usr/bin/tail -f /var/log/nginx/error.log

```

```

PS C:\Users\hungdg42304> curl localhost

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is
parsed.
RECOMMENDED ACTION:
Use the -UseBasicParsing switch to avoid script code execution.

Do you want to continue?

[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): y

StatusCode      : 200
StatusDescription : OK
Content          : <!DOCTYPE html>
                  <html>
                  <head>
                  <title>Welcome to nginx!</title>
                  <style>
                  html { color-scheme: light dark; }
                  body { width: 35em; margin: 0 auto;
                  font-family: Tahoma, Verdana, Arial, sans-serif; }
                  </style>...
RawContent       : HTTP/1.1 200 OK

```

```

hungdg@PC-BNKB167:~$ su - alice
Password:
alice@PC-BNKB167:~$ sudo systemctl restart nginx
alice@PC-BNKB167:~$ sudo systemctl status nginx
sudo: I'm sorry alice. I'm afraid I can't do that
alice@PC-BNKB167:~$ sudo tail -f /var/log/nginx/access.log
127.0.0.1 - - [22/Jun/2026:14:28:44 +0700] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; en-US
) WindowsPowerShell/5.1.26100.8115"

```

Phân quyền group webdev

```

hungdg@PC-BNKB167:~$ sudo chown root:webdev /var/www/html
hungdg@PC-BNKB167:~$ sudo chmod 2775 /var/www/html
hungdg@PC-BNKB167:~$ sudo -u alice touch /var/www/html/index.html
hungdg@PC-BNKB167:~$ ls -la /var/www/html/index.html
-rw-r--r-- 1 alice webdev 0 Jun 22 14:31 /var/www/html/index.html

```

Test user nagios

```

hungdg@PC-BNKB167:~$ sudo chown root:monitoring /var/log/nginx
hungdg@PC-BNKB167:~$ sudo chmod 750 /var/log/nginx
hungdg@PC-BNKB167:~$ sudo -u nagios ls /var/log/nginx/
access.log error.log
hungdg@PC-BNKB167:~$ sudo -u nagios cat /var/log/nginx/access.log | head -5
cat: /var/log/nginx/access.log: Permission denied

```

```

hungdg@PC-BNKB167:~$ sudo ls -l /var/log/nginx/
total 8
-rw-r----- 1 www-data adm 156 Jun 22 14:28 access.log
-rw-r----- 1 www-data adm 76 Jun 22 13:42 error.log

```

File log khi tạo ra không thuộc group monitor

Phân quyền lại 2 mục .log rồi kiểm tra lại

```

hungdg@PC-BNKB167:~$ sudo chown root:monitoring /var/log/nginx/access.log
hungdg@PC-BNKB167:~$ sudo chown root:monitoring /var/log/nginx/error.log
hungdg@PC-BNKB167:~$ ls -l /var/log/nginx/
ls: cannot open directory '/var/log/nginx/': Permission denied
hungdg@PC-BNKB167:~$ sudo ls -l /var/log/nginx/
total 8
-rw-r----- 1 root monitoring 156 Jun 22 14:28 access.log
-rw-r----- 1 root monitoring 76 Jun 22 13:42 error.log

```

```

hungdg@PC-BNKB167:~$ sudo -u nagios cat /var/log/nginx/access.log | head -5
127.0.0.1 - - [22/Jun/2026:14:28:44 +0700] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; en-US
) WindowsPowerShell/5.1.26100.8115"
hungdg@PC-BNKB167:~$ sudo -u nagios cat /var/log/nginx/error.log | head -5
2026/06/22 13:42:57 [notice] 6904#6904: using inherited sockets from "5;6;"

```

Backup tự động

```
hungdg@PC-BNKB167:~$ sudo mkdir /opt/scripts/  
hungdg@PC-BNKB167:~$ sudo vim /opt/scripts/auto_backup.sh  
hungdg@PC-BNKB167:~$ sudo chmod +x /opt/scripts/auto_backup.sh
```

```
#!/bin/bash  
# Production Backup Script  
  
set -euo pipefail # Dừng ngay nếu có lỗi  
  
BACKUP_ROOT='/var/backups/web'  
LOG='/var/log/backup.log'  
DATE=$(date '+%Y-%m-%d_%H%M')  
RETENTION=7 # Giữ backup 7 ngày  
  
log() { echo "[$(date '+%Y-%m-%d %H:%M:%S')] $*" | tee -a "$LOG"; }  
  
log '--- Bắt đầu backup ---'  
mkdir -p "$BACKUP_ROOT"  
  
# Backup web root  
tar -czf "$BACKUP_ROOT/webroot_${DATE}.tar.gz" /var/www/html 2>/dev/null  
log "Backup webroot: OK ($(du -sh $BACKUP_ROOT/webroot_${DATE}.tar.gz | cut -f1))"  
  
# Backup nginx config  
tar -czf "$BACKUP_ROOT/nginx_cfg_${DATE}.tar.gz" /etc/nginx/ 2>/dev/null  
log 'Backup nginx config: OK'  
  
# Xóa backup cũ  
DELETED=$(find "$BACKUP_ROOT" -mtime +${RETENTION} -name '*.tar.gz' -delete -print | wc -l)  
log "Đã xóa $DELETED file backup cũ hơn ${RETENTION} ngày"  
  
log '--- Backup hoàn thành ---'
```

Test thử scripts

```
hungdg@PC-BNKB167:~$ sudo /opt/scripts/auto_backup.sh  
[2026-06-22 15:01:57] --- Bắt đầu backup ---  
[2026-06-22 15:01:57] Backup webroot: OK (4.0K)  
[2026-06-22 15:01:57] Backup nginx config: OK  
[2026-06-22 15:01:57] Đã xóa 0 file backup cũ hơn 7 ngày  
[2026-06-22 15:01:57] --- Backup hoàn thành ---  
hungdg@PC-BNKB167:~$  
cat /var/log/backup.log  
[2026-06-22 15:01:57] --- Bắt đầu backup ---  
[2026-06-22 15:01:57] Backup webroot: OK (4.0K)  
[2026-06-22 15:01:57] Backup nginx config: OK  
[2026-06-22 15:01:57] Đã xóa 0 file backup cũ hơn 7 ngày  
[2026-06-22 15:01:57] --- Backup hoàn thành ---  
hungdg@PC-BNKB167:~$
```


Tạo cronjob

```
hungdg@PC-BNKB167:~$ sudo crontab -e
no crontab for root - using an empty one
Select an editor. To change later, run select-editor again.
 1. /bin/nano      <---- easiest
 2. /usr/bin/vim.basic
 3. /usr/bin/vim.tiny
 4. /bin/ed

Choose 1-4 [1]: 3
crontab: installing new crontab
```

```
0 3 * * * /opt/scripts/auto_backup.sh
0 6 * * 1 /usr/bin/find /tmp -mtime +7 -delete 2>/dev/null
```

Bảo mật SSH

```
> EOF
Port 2222
PermitRootLogin no
PasswordAuthentication no
PubkeyAuthentication yes
MaxAuthTries 3
MaxSessions 5
ClientAliveInterval 300
ClientAliveInterval 300
ClientAliveCountMax 2
AllowGroups webdev
Banner /etc/ssh/banner.txt
```

```
> EOF
#####
#   HỆ THỐNG BỊ GIÁM SÁT – CHỈ TRUY CẬP HỢP LỆ #
#####
```

test port 2222

```
hungdg@PC-BNKB167:~$ ss -tlnp | grep 2222
LISTEN 0      128          0.0.0.0:2222  0.0.0.0:*
LISTEN 0      128          [::]:2222  [::]:*
```

Root bị chặn

```

hungdg@PC-BNKB167:~$ ssh root@localhost -p 2222
The authenticity of host '[localhost]:2222 ([127.0.0.1]:2222)' can't be established.
ED25519 key fingerprint is: SHA256:G6++QGQ67zCGd0Dd281Zu58SmLbPXUqEfVTLGAwjEgc
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? ye
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '[localhost]:2222' (ED25519) to the list of known hosts.
#####
#   HỆ THỐNG BỊ GIÁM SÁT – CHỈ TRUY CẬP HỢP LỆ #
#####

root@localhost: Permission denied (publickey).

```

Test ssh bằng alice user:

```

hungdg@PC-BNKB167:~$ ssh -i /home/hungdg/.ssh/id_ed25519 -p 2222 alice@172.18.236.53
The authenticity of host '[172.18.236.53]:2222 ([172.18.236.53]:2222)' can't be established.
ED25519 key fingerprint is: SHA256:G6++QGQ67zCGd0Dd281Zu58SmLbPXUqEfVTLGAwjEgc
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[172.18.236.53]:2222' (ED25519) to the list of known hosts.
#####
#   HỆ THỐNG BỊ GIÁM SÁT – CHỈ TRUY CẬP HỢP LỆ #
#####

Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.114.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Jun 22 15:33:12 +07 2026

System load:  0.0               Processes:    51
Usage of /:   0.2% of 1006.85GB  Users logged in:  0
Memory usage: 6%               IPv4 address for eth0: 172.18.236.53
Swap usage:  0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

alice@PC-BNKB167:~$ |

```

```

hungdg@PC-BNKB167:~$ ssh -p 2222 alice@172.18.236.53
#####
#   HỆ THỐNG BỊ GIÁM SÁT - CHỈ TRUY CẬP HỢP LỆ #
#####

Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.6.114.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Jun 22 16:23:56 +07 2026

System load:  0.0               Processes:            55
Usage of /:   0.2% of 1006.85GB Users logged in:      0
Memory usage: 6%               IPv4 address for eth0: 172.18.236.53
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Mon Jun 22 15:33:14 2026 from 172.18.236.53
alice@PC-BNKB167:~$ |

```

Alice restart nginx được

```

hungdg@PC-BNKB167:~$ su - alice
Password:
alice@PC-BNKB167:~$ sudo systemctl restart nginx
alice@PC-BNKB167:~$ sudo systemctl status nginx
sudo: I'm sorry alice. I'm afraid I can't do that
alice@PC-BNKB167:~$ sudo tail -f /var/log/nginx/access.log
127.0.0.1 - - [22/Jun/2026:14:28:44 +0700] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; en-US
) WindowsPowerShell/5.1.26100.8115"
|

```

SGID trên web root

```

hungdg@PC-BNKB167:~$ ls -la /var/www/html
total 12
drwxrwsr-x 2 root webdev 4096 Jun 22 14:31 .
drwxr-xr-x 3 root root   4096 Jun 22 13:42 ..
-rw-r--r-- 1 alice webdev    0 Jun 22 14:31 index.html
-rw-r--r-- 1 root root      615 Jun 22 13:42 index.nginx-debian.html

```

Backup chạy và tạo file

```

hungdg@PC-BNKB167:~$ ls -la /var/backups/web/
total 20
drwxr-xr-x 2 root root 4096 Jun 22 15:01 .
drwxr-xr-x 3 root root 4096 Jun 22 15:01 ..
-rw-r--r-- 1 root root 6913 Jun 22 15:01 nginx_cfg_2026-06-22_1501.tar.gz
-rw-r--r-- 1 root root 598 Jun 22 15:01 webroot_2026-06-22_1501.tar.gz

```

Cronjob đã đăng ký

```

hungdg@PC-BNKB167:~$ sudo crontab -l
0 3 * * * /opt/scripts/auto_backup.sh
0 6 * * 1 /usr/bin/find /tmp -mtime +7 -delete 2>/dev/null

```

Demo đăng nhập bằng user alice tạo 1 file .txt xem thử group = webdev hay không?

```

alice@PC-BNKB167:~$ ls -l /var/www/html
total 4
-rw-r--r-- 1 alice webdev 0 Jun 22 14:31 index.html
-rw-r--r-- 1 root root 615 Jun 22 13:42 index.nginx-debian.html
alice@PC-BNKB167:~$ touch /var/www/html/test_alice.txt
alice@PC-BNKB167:~$ ls -l /var/www/html
total 4
-rw-r--r-- 1 alice webdev 0 Jun 22 14:31 index.html
-rw-r--r-- 1 root root 615 Jun 22 13:42 index.nginx-debian.html
-rw-rw-r-- 1 alice webdev 0 Jun 22 15:48 test_alice.txt
alice@PC-BNKB167:~$

```